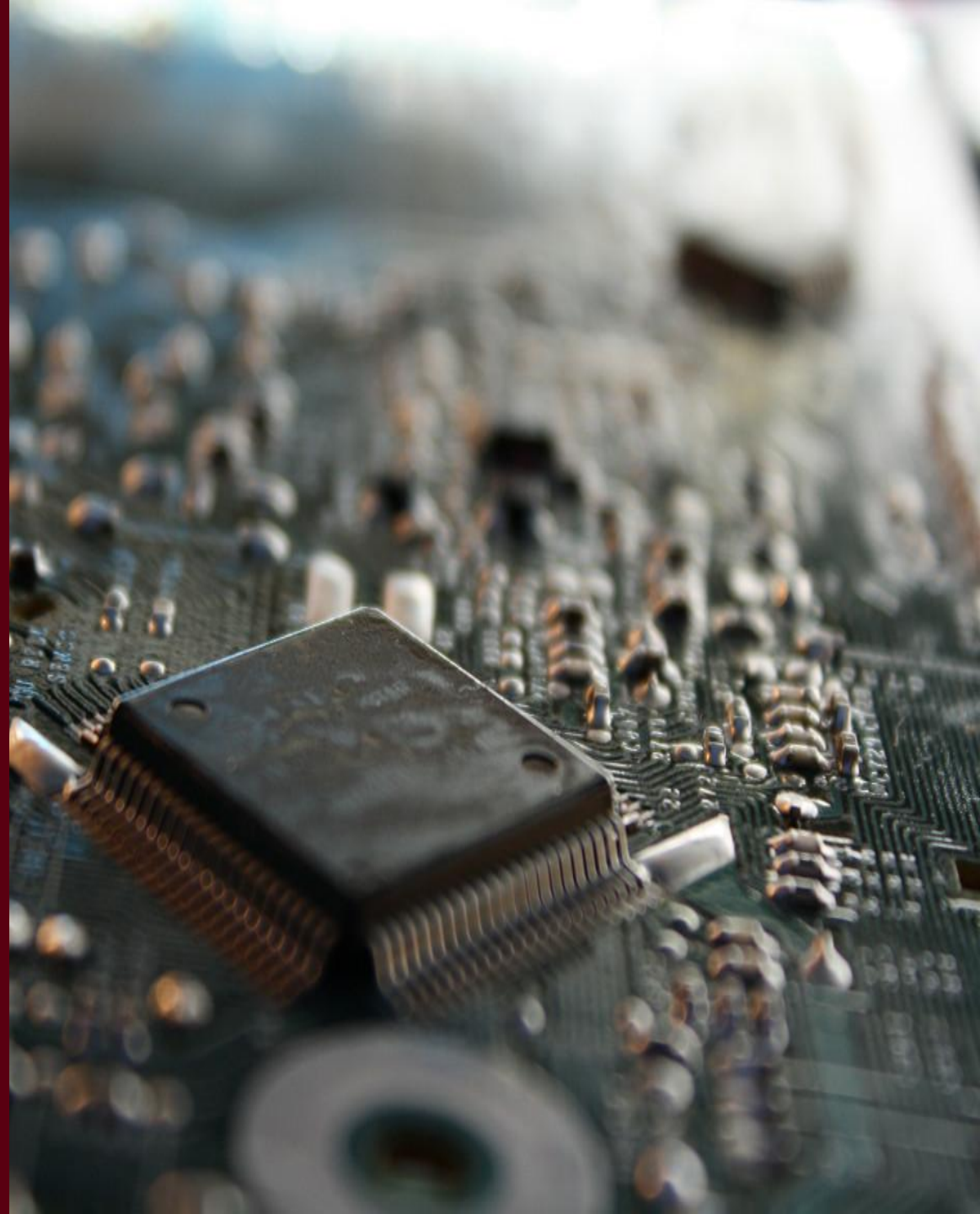


Agentic AI For Semiconductor Design and Manufacturing

From Wafer Data Workflows to Intelligent EDA
Systems

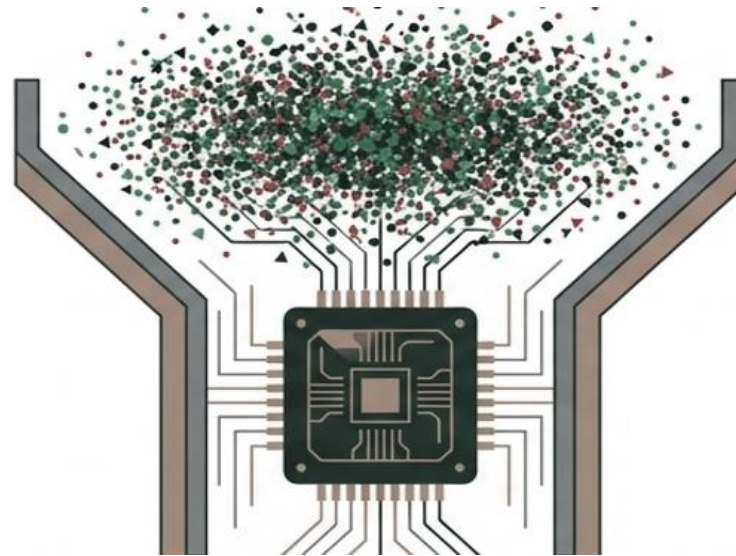
Janhavi Giri, PhD
Principal Architect, NetApp
09/17/2026



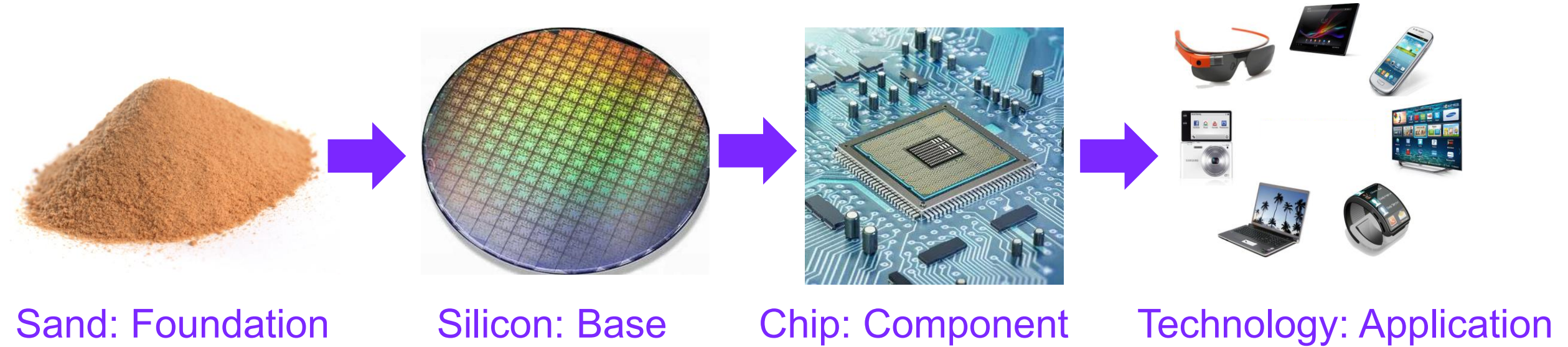
Outline

- Introduction: Semiconductor Industry & Data Explosion
- Evolution from ML to Agentic AI
- AI Agents: Components, Logic & Tool Integration
- Case Studies: Design, Verification, & Manufacturing
- System Design: Data Models, Storage & Compute
- Wafer Yield Inspection AI Agent Demo
- Performance, Scalability & Cost
- Advanced Extensions
- Roadmap & Actionable Strategies
- Key takeaway

Semiconductor Industry & Data Deluge

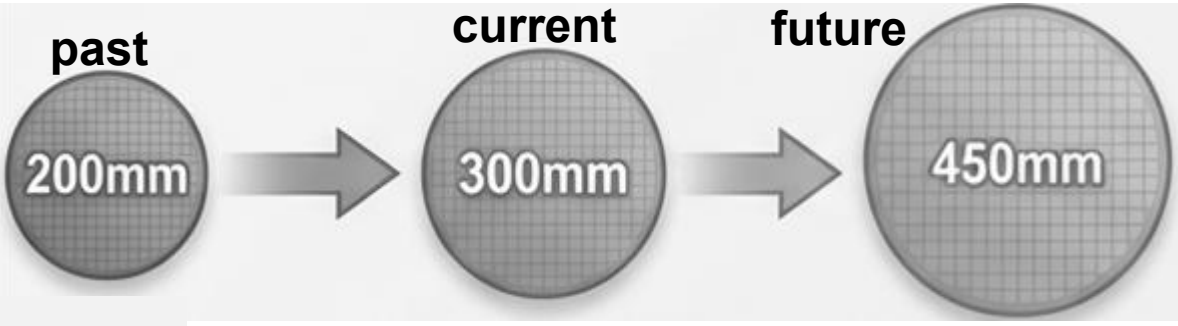


From Sand to Silicon: The Pillar of Our Digital World

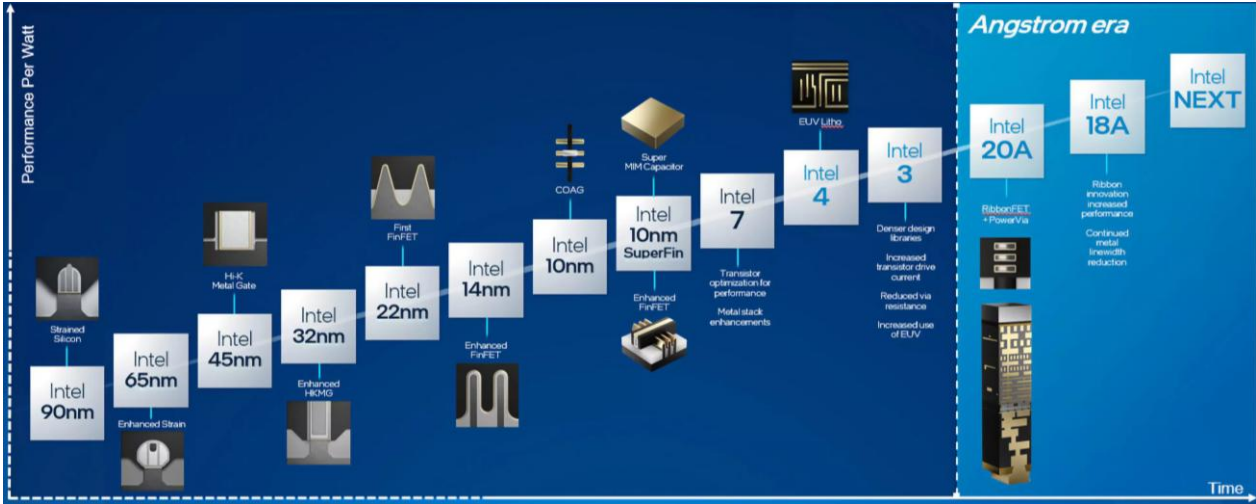


Semiconductors: Invisible nervous system of global civilization, embedded in nearly every facet of modern life

Scaling And Complexity Drivers: Trillion Dollar Semiconductor Industry

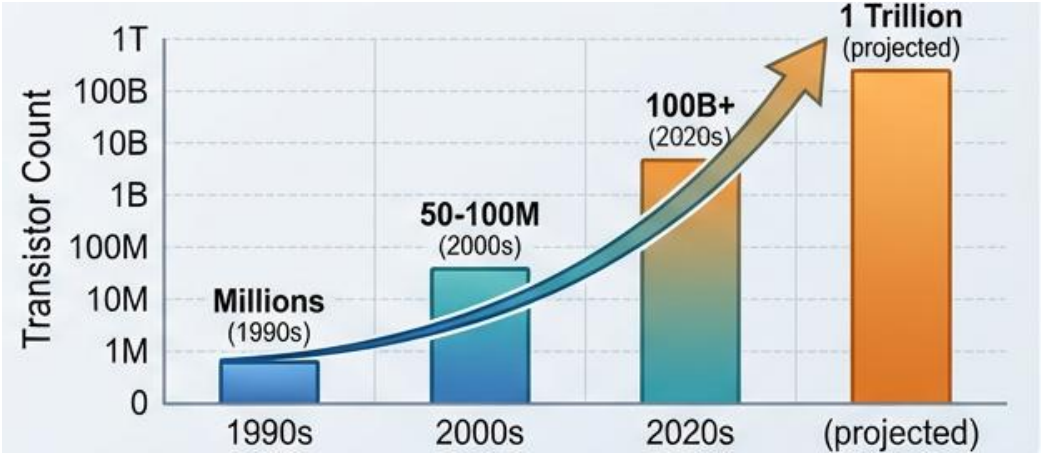


Wafer Size Evolution: 50% increase per decade

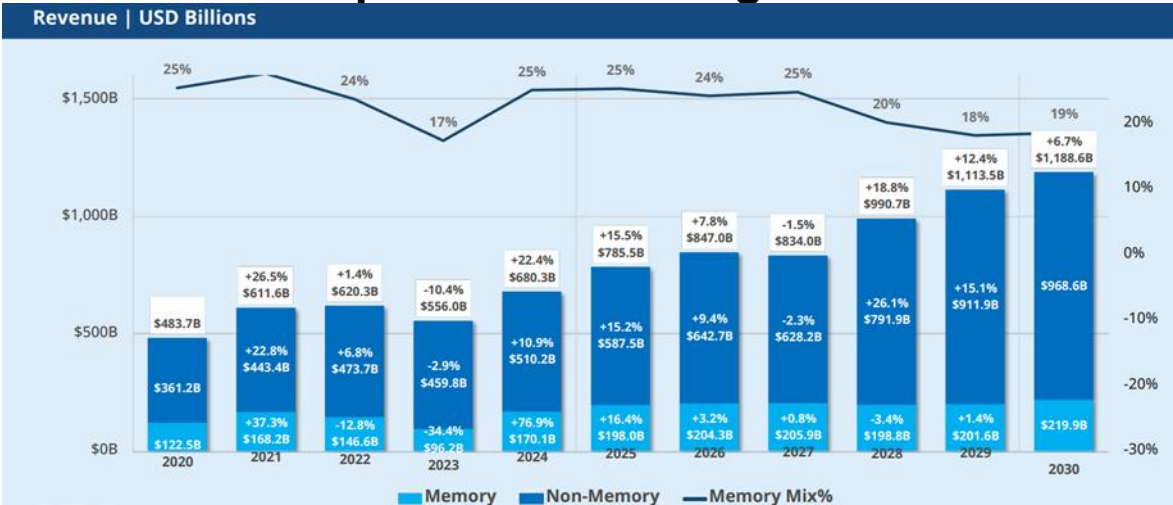


Intel Process Technology: Node Size Reduction (Moore’s law)

<https://www.techspot.com/article/1856-aiming-for-atoms-chip-manufacturing/>, <https://www.asml.com/en/technology/all-about-microchips/microchip-basics>, www.idc.com



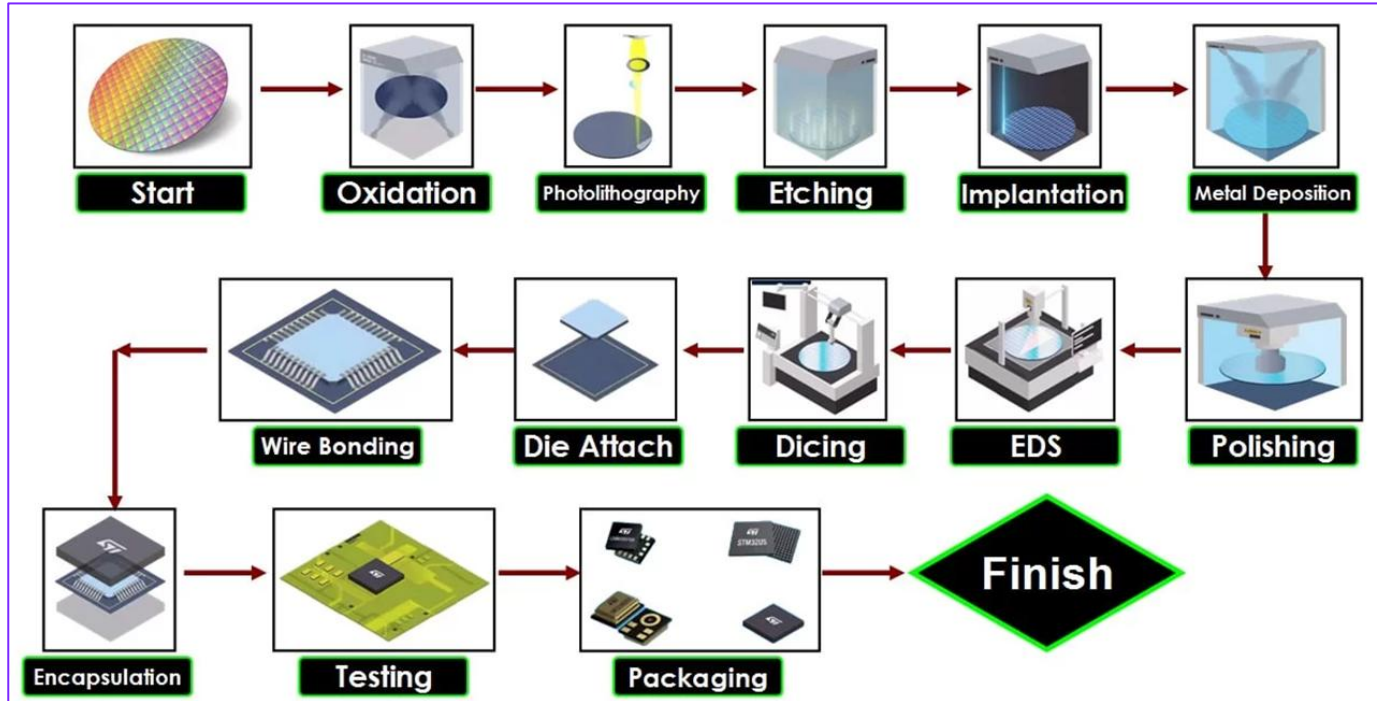
Transistor Count: Trillions packed on a microchip “size of our fingernail”



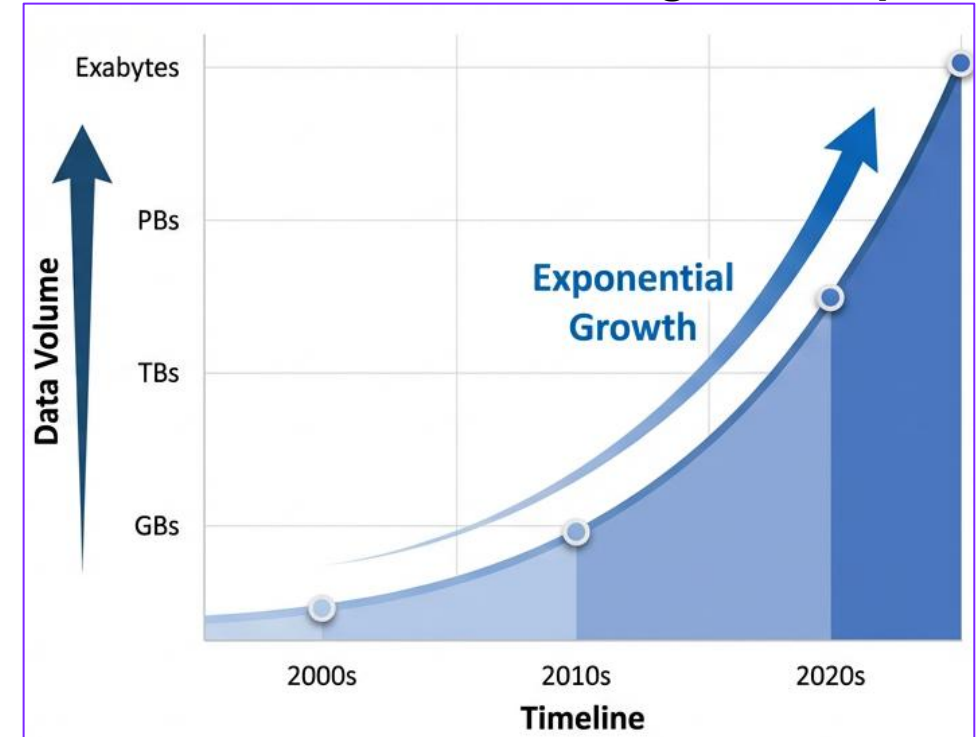
WW Semiconductor Market 2030 Outlook (AI-Chip demand surge)

The Foundation of Innovation: From Raw Silicon to Global Connectivity

Semiconductor Manufacturing: High Precision, 1000+ steps



Semiconductor Manufacturing Data Explosion



Data Sources: Equipment sensors, Metrology systems, Wafer inspection tools, Test data systems, Mask data storage

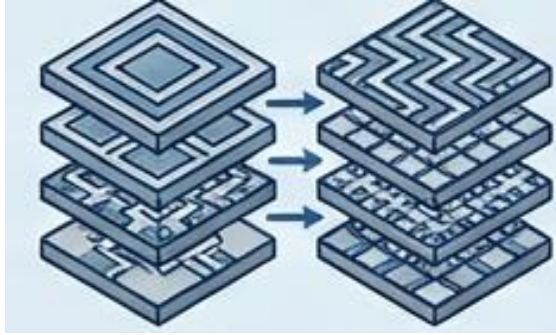
Terabytes of data generation daily

Data Deluge: 50+ TB/day per fab; >10 PB/year : 25,000× increase since 1996

Semiconductor Manufacturing Challenges



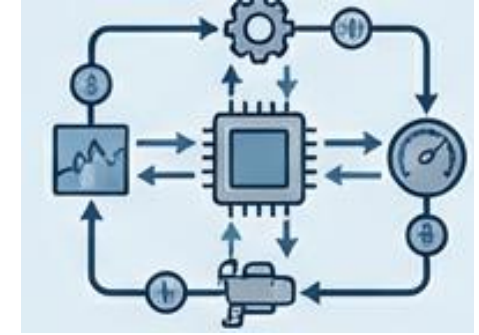
1000+ Process Steps



Multi-Patterning



RET/OPC (Mask data)



Advanced Process Control (APC)



Predictive Maintenance Analytics



Virtual Metrology & Yield Prediction

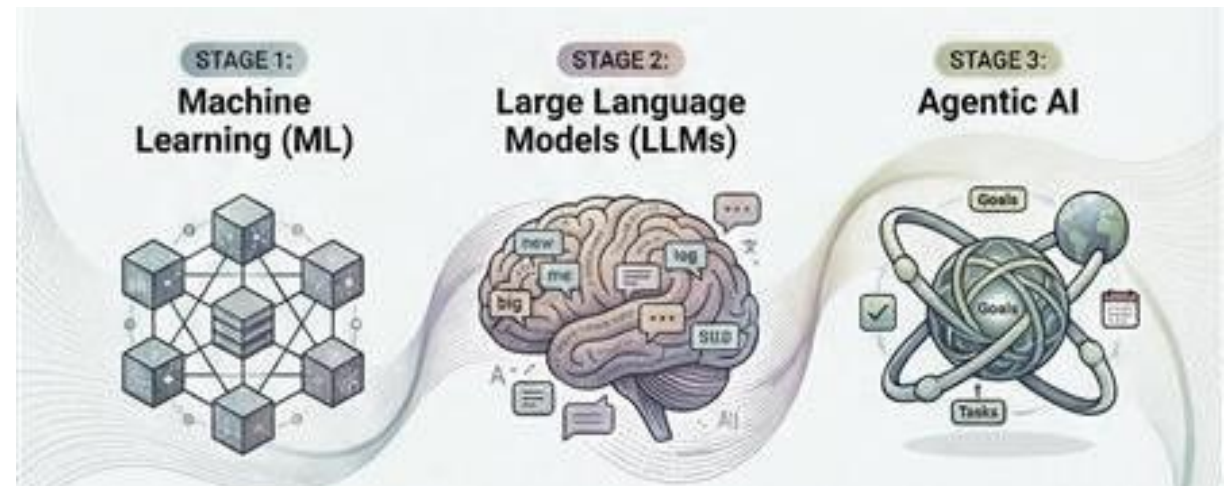


Real-Time Equipment Monitoring



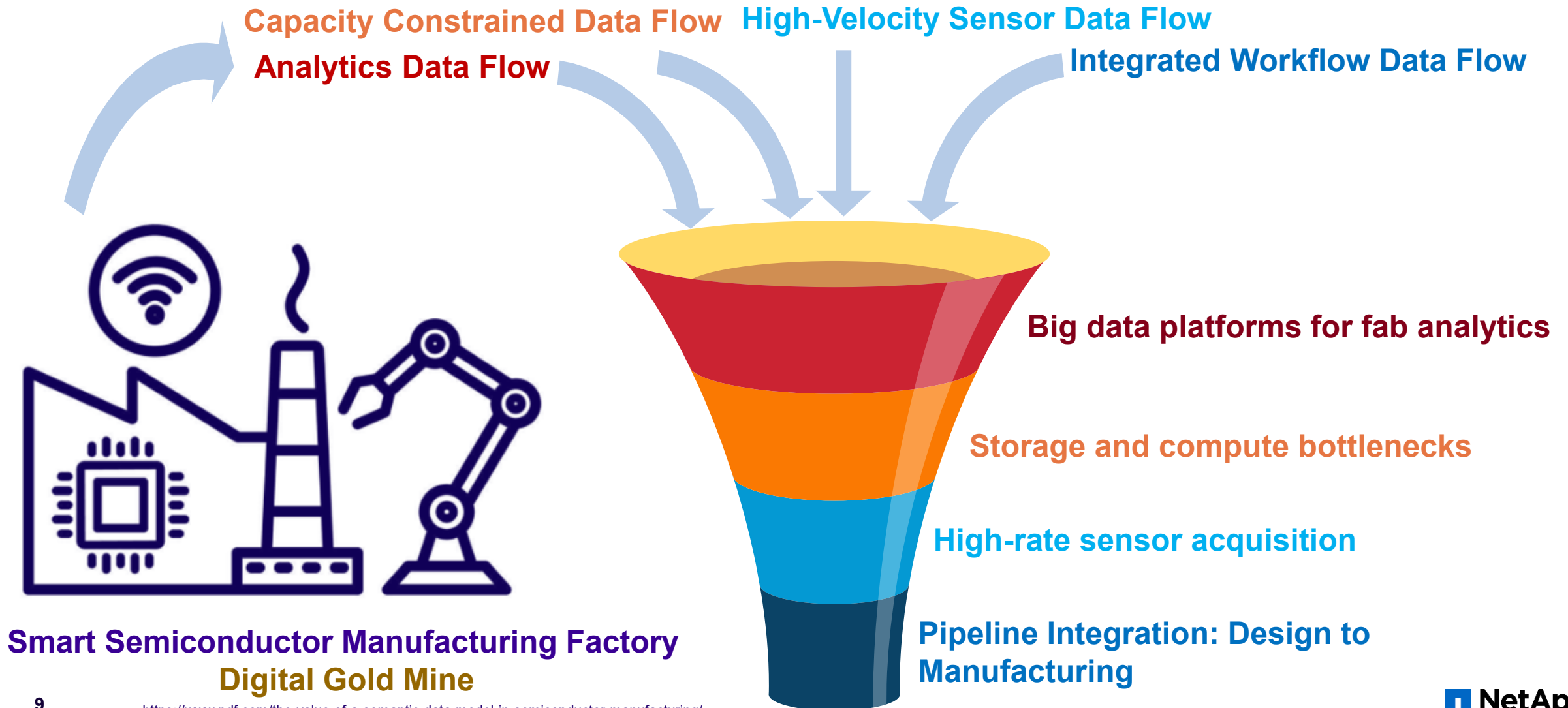
Fault Detection Systems

Evolution from ML to Agentic AI



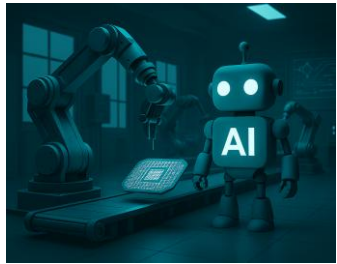
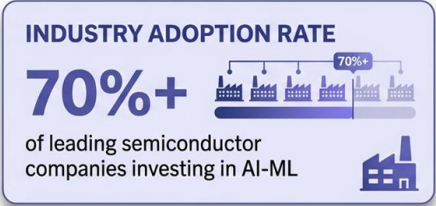
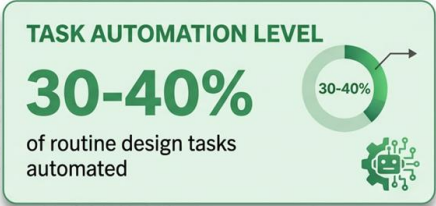
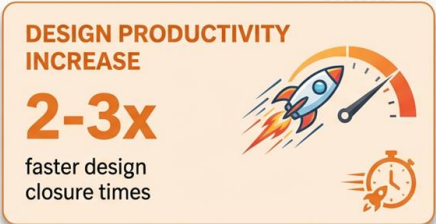
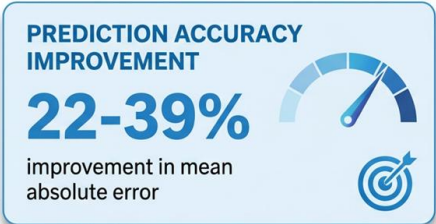
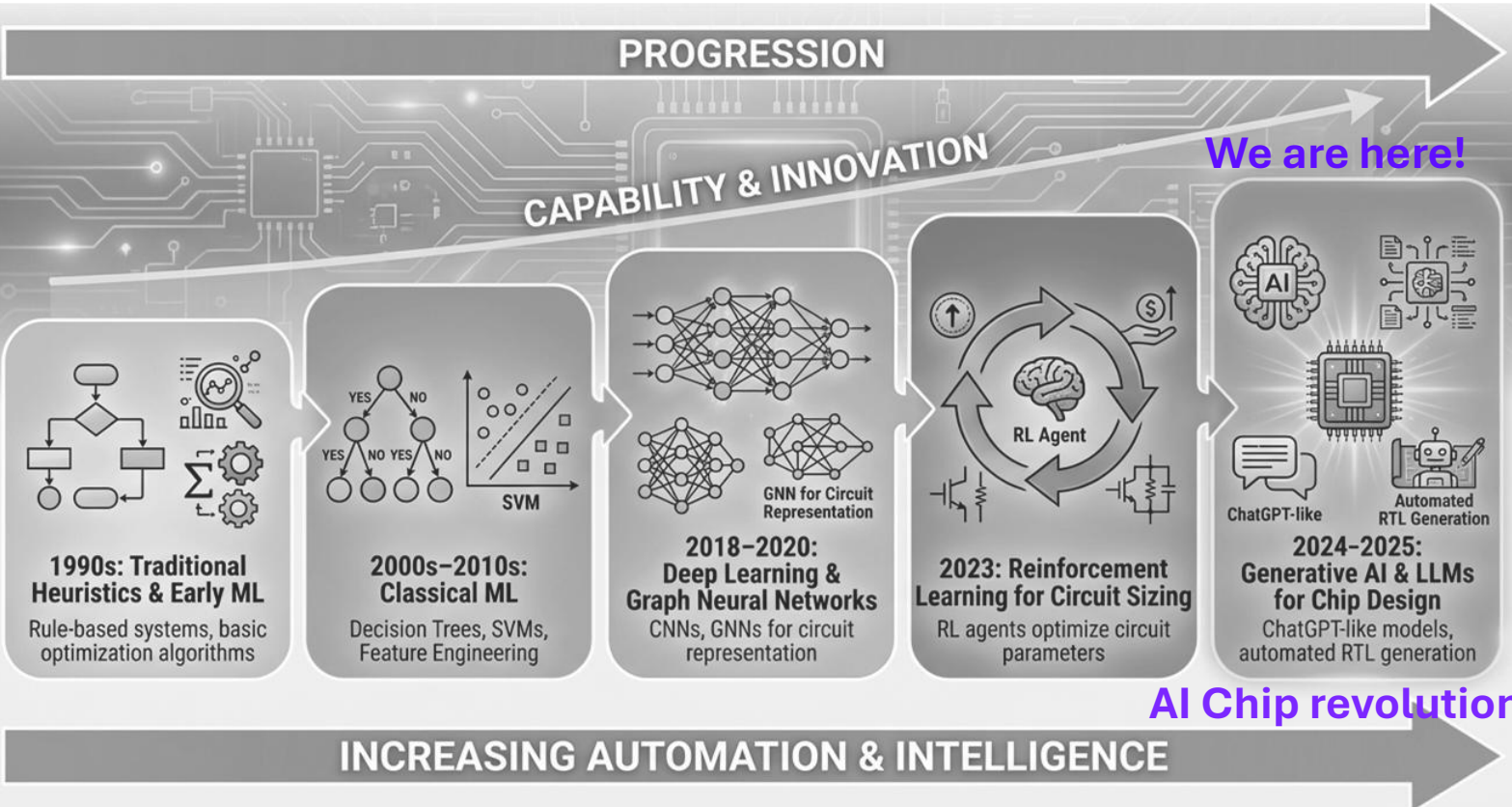
Data Intensive Semiconductor Manufacturing Landscape

1 minute of a gigafab manufacturing process comprises: 75,000 wafer move events, 15,000 sensor readings, 9.6 GB of EUV data collected, 95 GB of equipment data collected, and 360,000 recipe step events!



Evolution of AI-ML In Semiconductor Industry

The Journey from Heuristics to Generative AI with Key Impact Metrics

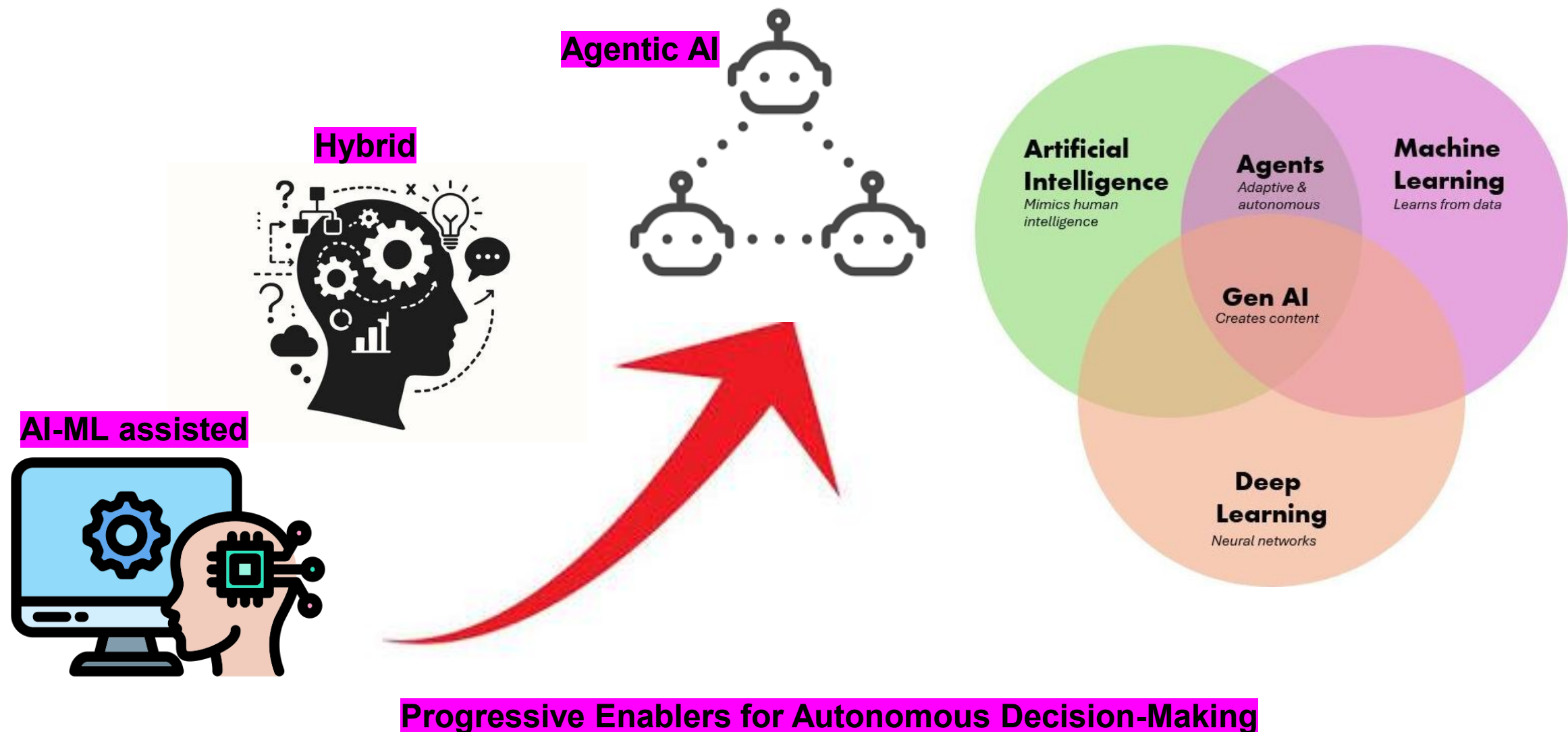


Agentic AI: Dawn of Autonomous Intelligence

Three decades of AI-ML evolution fundamentally transform semiconductor design and manufacturing from manual processes to intelligent automation.

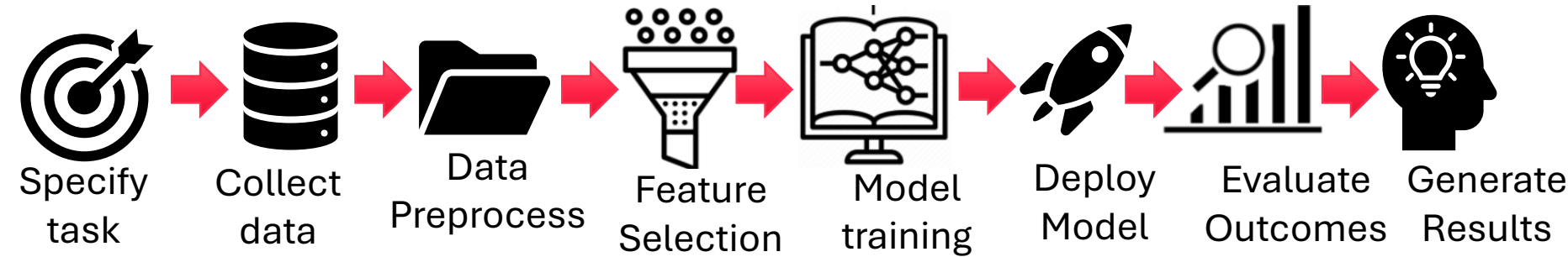
From Assistance to Autonomy: The Agentic AI Leap

Transforming Semiconductor Workflows with Self-Directed Intelligence



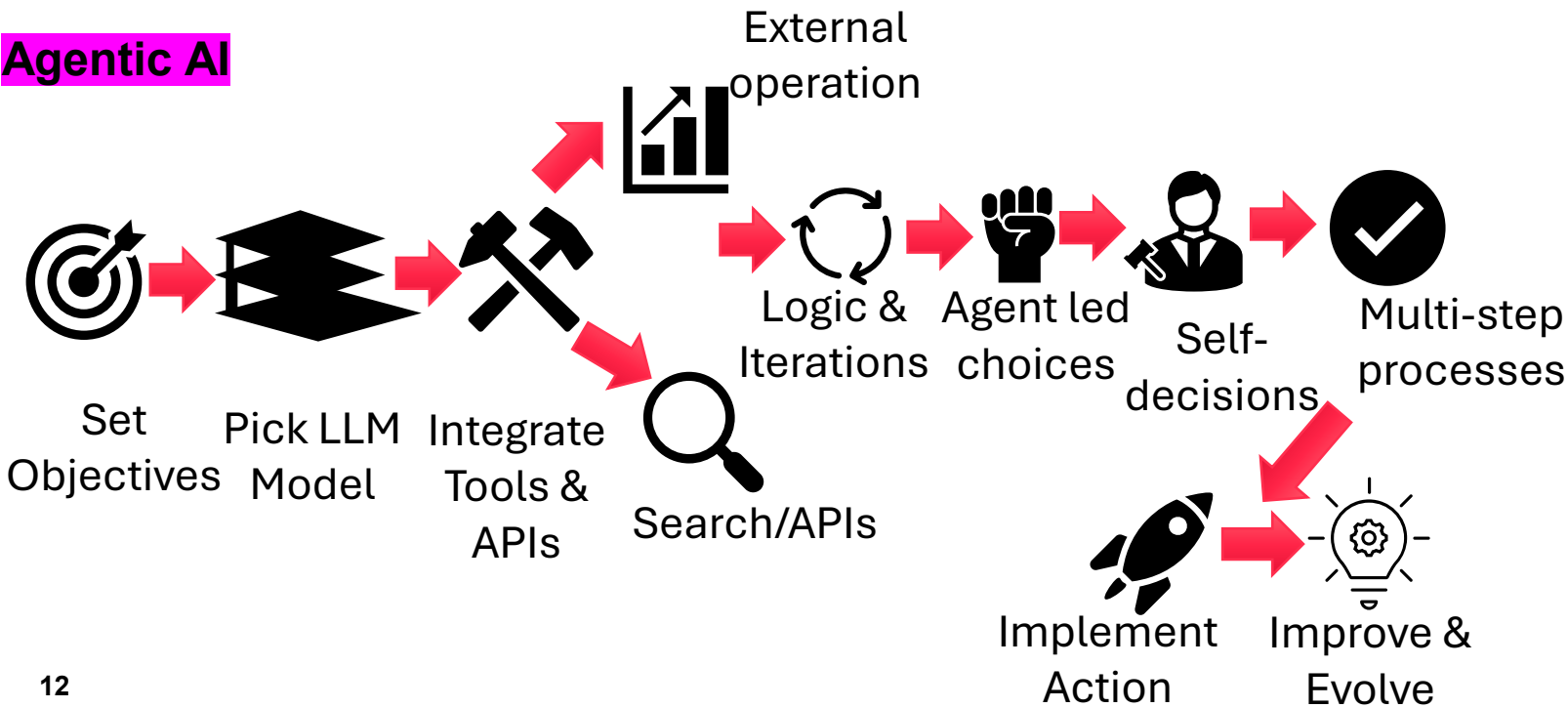
Beyond Scripts: The Power of Autonomous AI Agents (The “Why”)

Traditional analytics



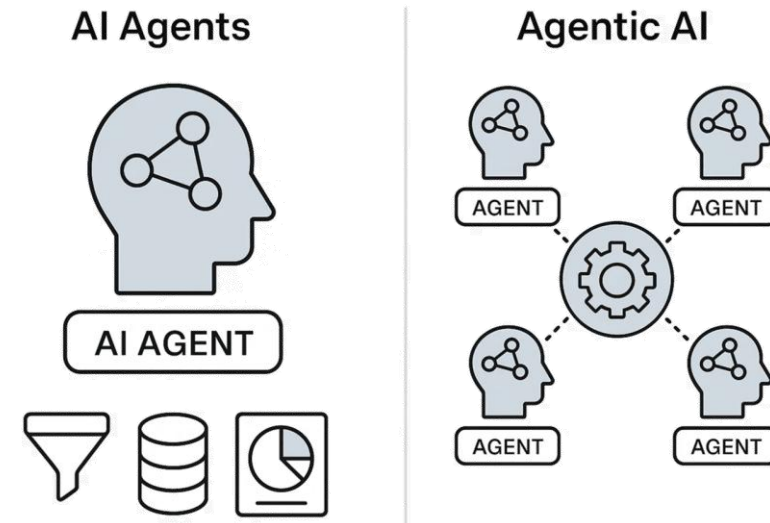
Traditional analytics workflows are rigid , fixed thresholds, static rules, and lots of human oversight

Agentic AI

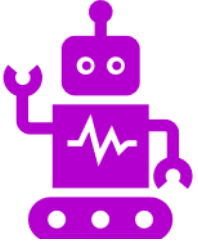


Agentic AI brings democratization of expertise, accelerated time to market, consistent quality and scalability

AI Agents: Components, Logic & Tool Integration

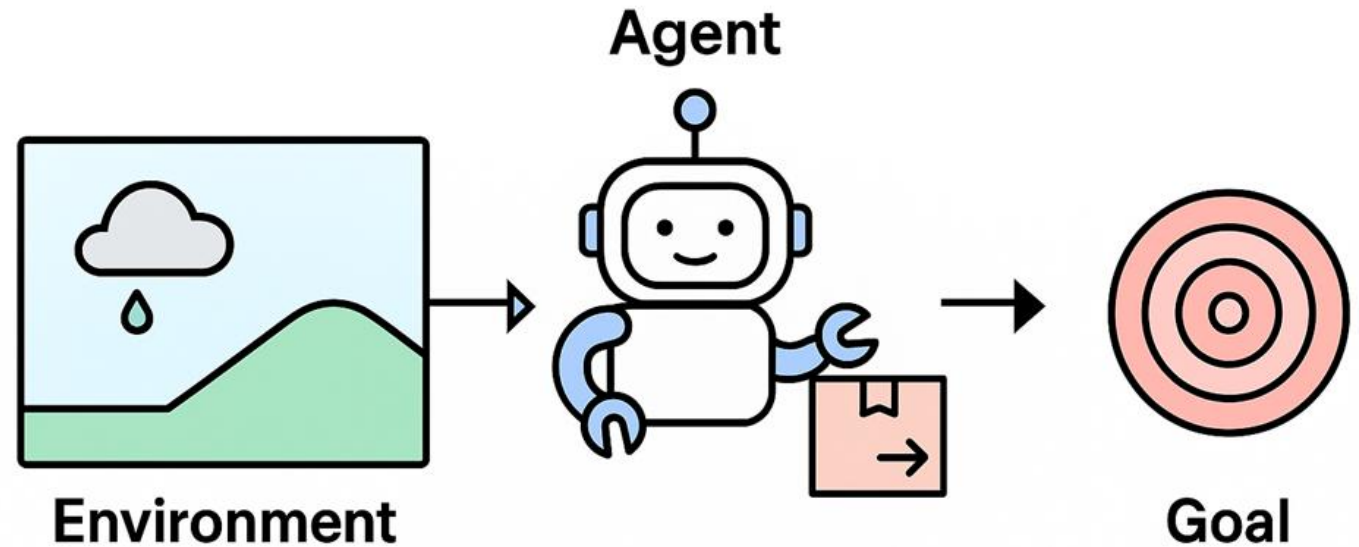


What is an AI Agent?



An agent is a system that perceives its environment, makes decisions, and takes actions to accomplish defined goals.

- Software engineering AI agents
- AI voice & phone call agents
- General-purpose AI research agents
- AI data analysis agents
- Computer use & UI navigation agents

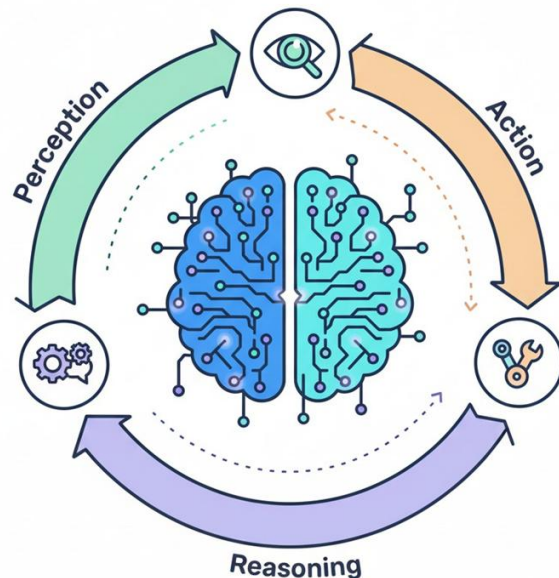


Key Components of AI Agents and the three pillars powering them



Components

- **Perception:** Sensing environment and inputs (ex. Text, images, sensors)
- **Reasoning & Planning:** Decision-making, task decomposition, goal setting
- **Action:** Executing tasks via tools or direct effectors

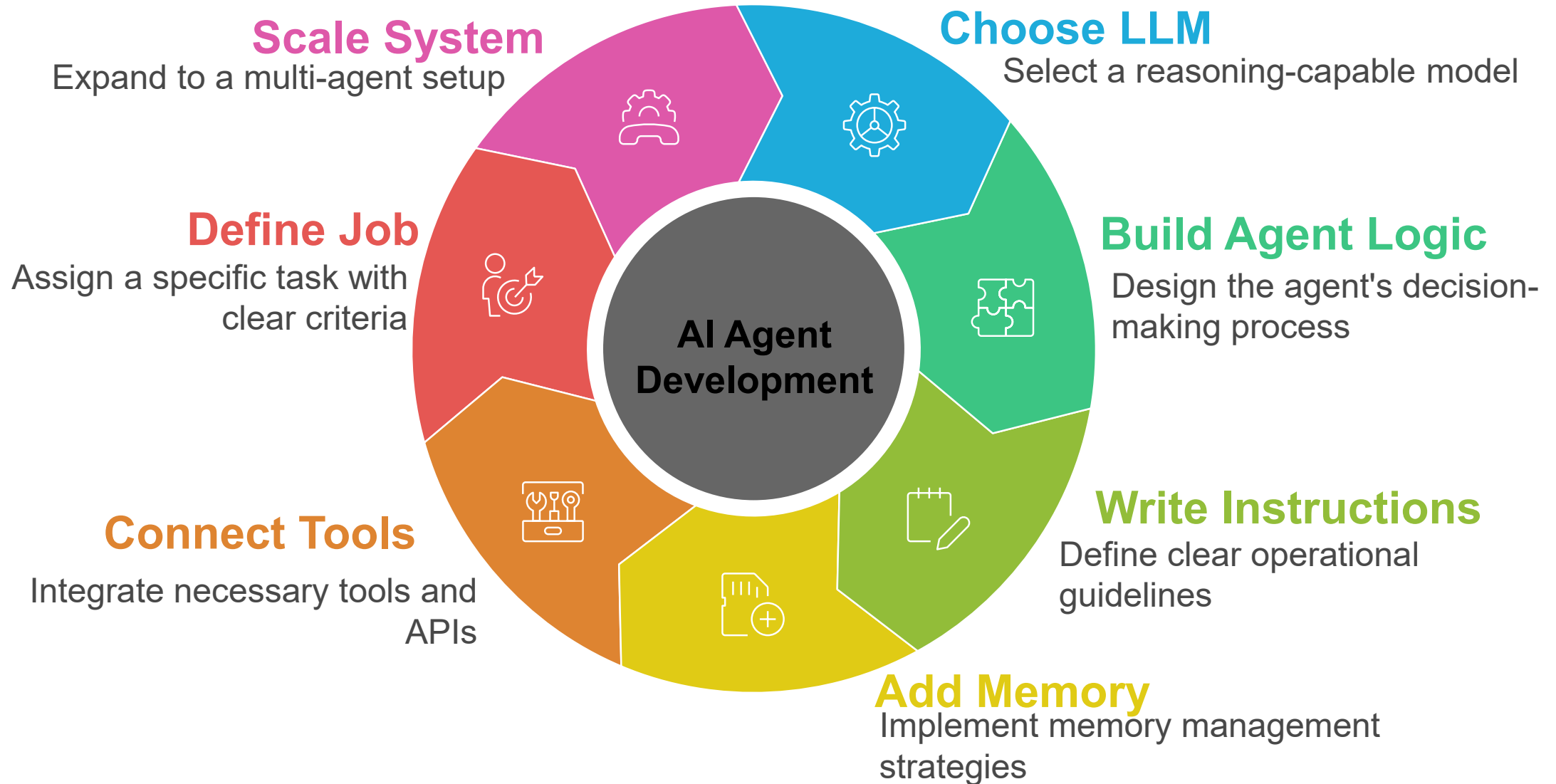


Pillars

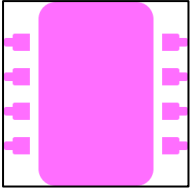
- **Planning:** LLMs enable adaptive problem-solving in AI agents. (**“the Brain”**)
- **Tools Integration:** External tools enhance AI agent capabilities. (**“the Hands and Eyes”**)
- **Memory:** Short-term and long-term memory support AI learning.



AI Agent Building Blocks



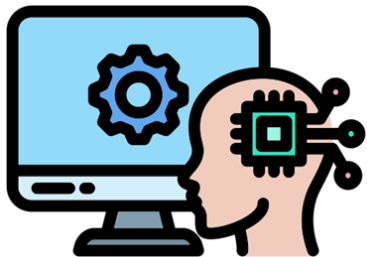
What is Agentic AI?



AI systems designed to behave with agency—they plan, reason, adapt, and coordinate actions autonomously over time (design paradigm for autonomous systems).

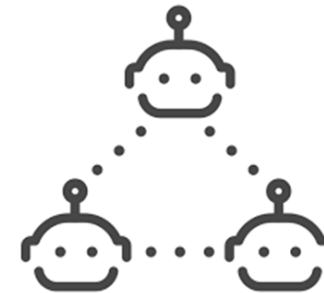
Traditional “Narrow” AI

- Pattern recognition on isolated tasks
- Flags timing violations or defects
- Requires predefined parameters
- Single-model, single-dataset scope

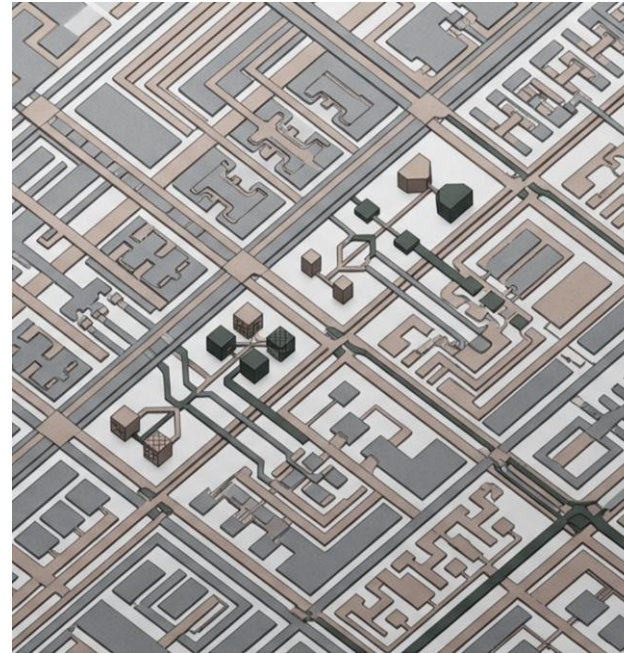


Agentic AI

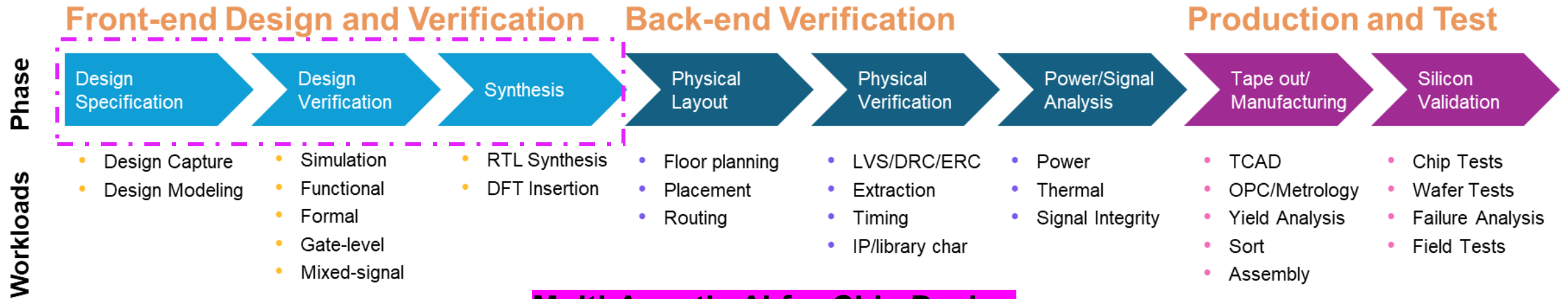
- Chain-of-thought reasoning + tool use
- Diagnoses root cause, scripts fixes, runs regression
- Goal-oriented with adaptive planning
- Multi-agent orchestration across workflows



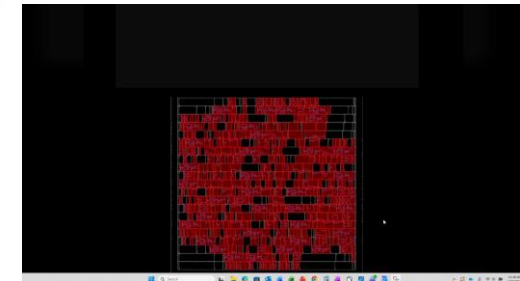
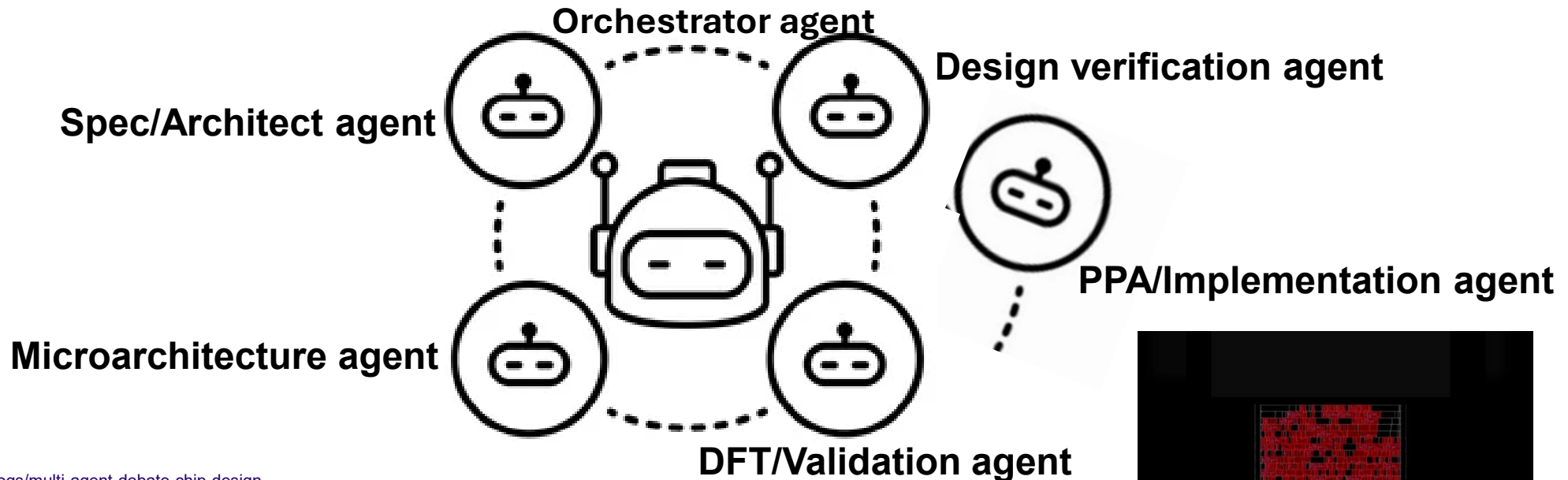
Case Studies: Design, Verification & Manufacturing



Multi-Agent AI for EDA: IC Chip Design to Verification

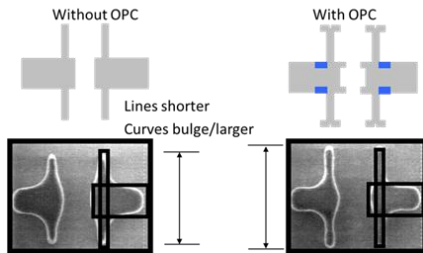
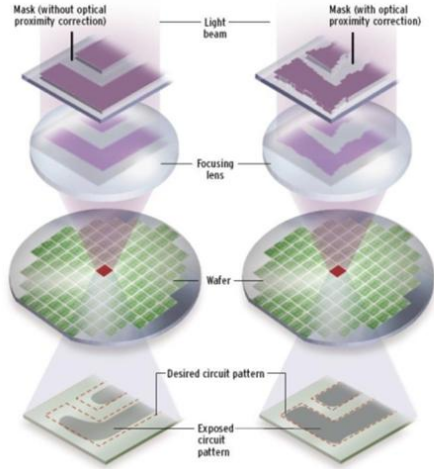


Multi-Agent AI for Chip Design



AI Agent for IC Manufacturing: OPC (Optical Proximity Correction) Manufacturing Aware Optimization

OPC one of RETs compensating for image errors induced due to light diffraction or lithography errors

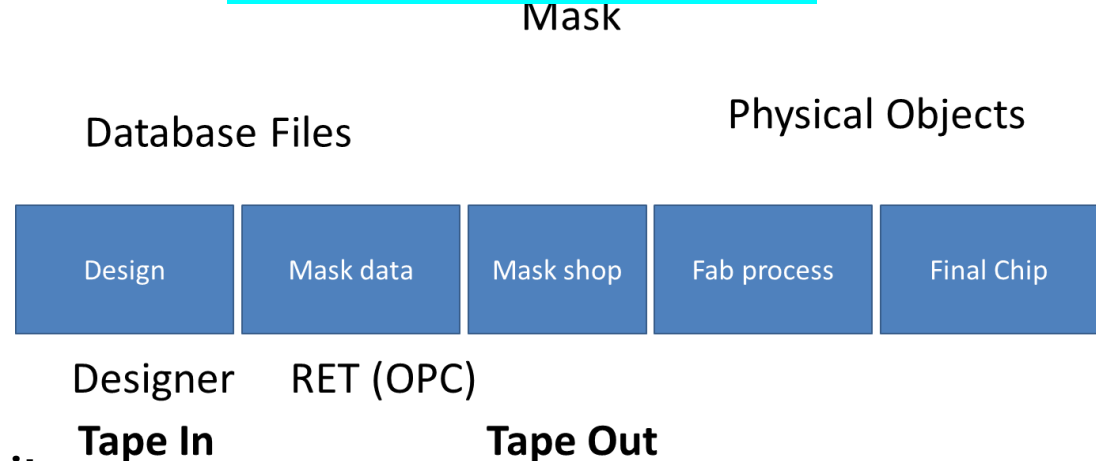


Rayleigh Equation

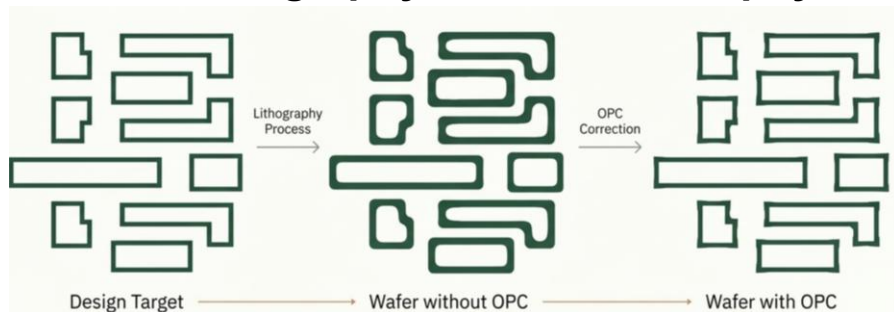
$$CD_{min} = k_1 \frac{\lambda}{NA}$$

Resolution limit

Traditional OPC Workflow



Nanoscale lithography causes severe physical distortions

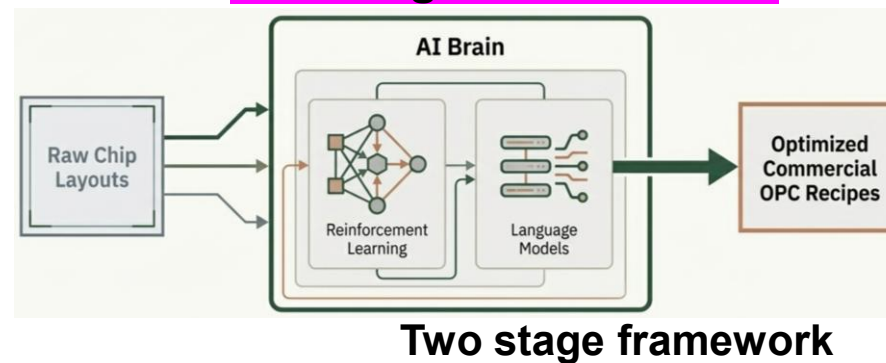


OPC requires: Universal engine (OPC engine) & a bespoke recipe (OPC recipe)

Recipe development: Manual human trial & error

Chen, G., Wang, S., Yu, B., & Ren, H. (2024). Intelligent OPC Engineer Assistant for Semiconductor Manufacturing. arXiv preprint arXiv:2408.12775

OPC Engineer Assistant

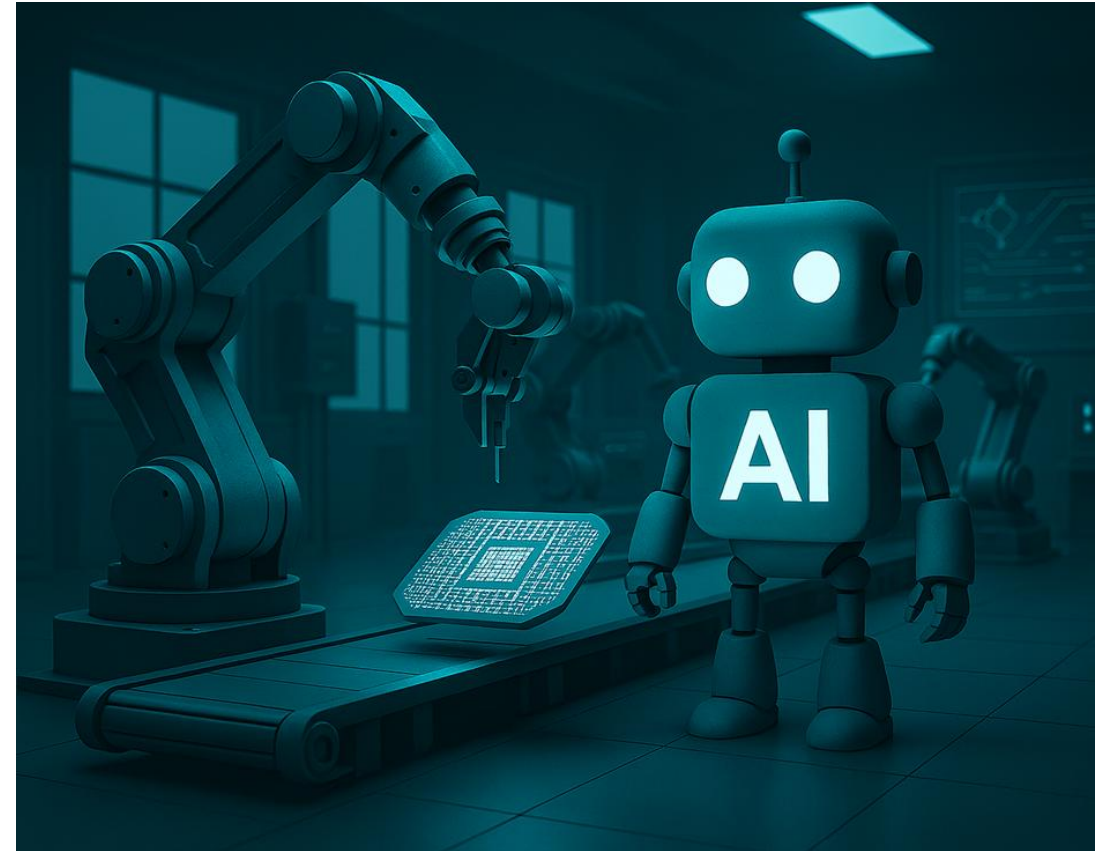


1. **RL** for optimal OPC parameters
2. **Multi-modal LLMs** for rule-based recipe through feature labeling and decision tree construction

- Reduces EPE & PVB by > 10% (no runtime impact)
- Automates full-time effort of experienced OPC engineers

Chip Design EDA Agents in Action

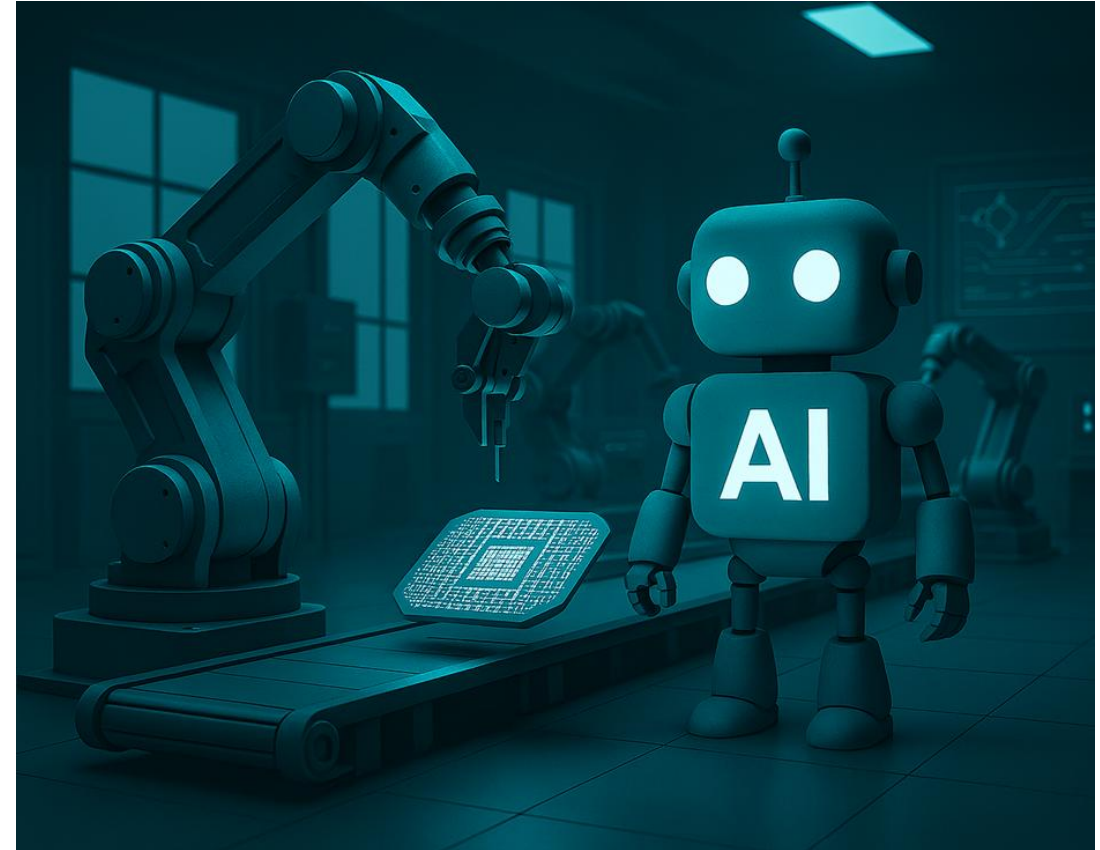
- **Cadence — ChipStack AI "Super" Agent**
 - Sub-agents for IP design, verification, signoff, debug, and SoC layout
 - Ingests specs to form mental model, generates + executes test code
 - Claims up to 10x productivity boost; adopted by Qualcomm, NVIDIA, Altera
- **Siemens EDA — Solido Generative & Agentic AI**
 - Natural language triggers EDA workflows end-to-end
 - Covers schematic capture, simulation, library characterization, layout
 - Integrated with NVIDIA NIM + Llama Nemotron for real-time orchestration
- **ChipAgents RCA — Autonomous Bug Resolution**
 - Multi-agent prover-verifier algorithm for waveform + code analysis
 - 12x speedup vs. human engineer on complex protocol bugs (AXI, PCIe)
 - Deployed at 50+ semiconductor companies including top-20 globally



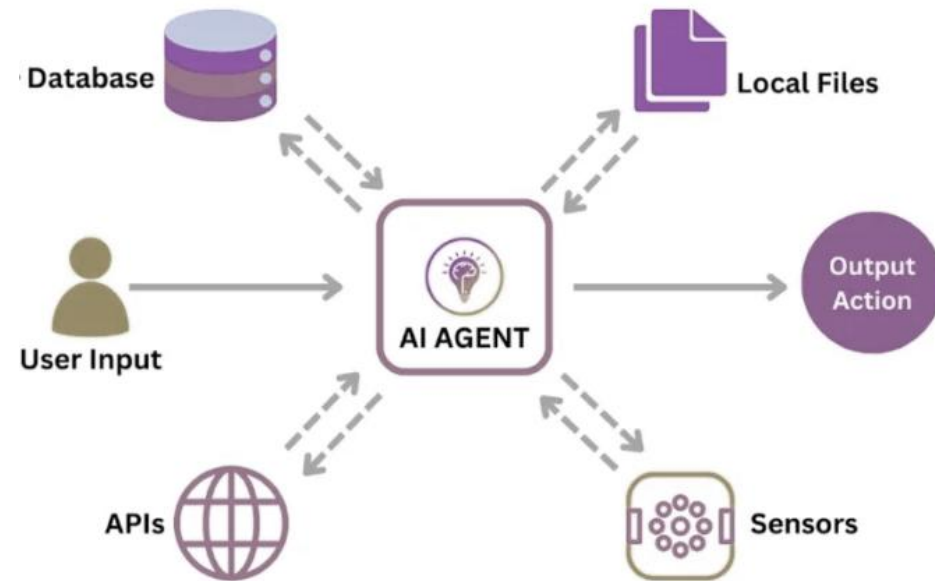
The Register, Feb 2026, DAC 2025 / Siemens Newsroom, Jun 2025, ChipAgents Blog, Nov 2025 (All case studies sourced from public announcements and published reports)

Manufacturing Yield & Root Cause Analysis in Action

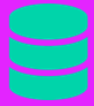
- **Intel — AI-Driven Gross Failure Area Detection**
 - Deep learning + ML models auto-detect GFA patterns on end-of-line wafers
 - Deployed across Virtual Factory network — patterns shared globally across fabs
 - 2,500+ wafers tagged/week with 16 models in production (scaled since 2021)
- **PDF Solutions — Semantic, Agentic & Secure Platform**
 - Semiconductor-native knowledge graphs anchor LLM reasoning (lot → wafer → die → package)
 - Model Context Protocol enables autonomous workflow execution from ingestion to plot generation
 - Air-gapped on-prem LLM option for IP sovereignty; engineers retain full transparency
- **Lam Research — Fabtex™ Yield Optimizer**
 - AI/ML on virtual silicon digital twins to recommend inline metrology target changes
 - Addresses multi-step process interactions in complex 3D structures
 - Saves fabs multiple weeks to achieve target yields with significant cost savings



Designing AI Agents



System Design: Data Models, Storage and Compute



Data Models

- Semiconductor-native knowledge graphs encoding fab hierarchy
- Multimodal data lake: wafer maps, metrology, equipment logs, test results
- Domain ontology eliminates LLM hallucinations on fab terminology



Storage Infrastructure

- High-throughput object & file storage for 50+ TB/day ingestion
- Data versioning & lineage for ML reproducibility (*FlexClone paradigm*)
- Air-gapped on-prem options for IP sovereignty in advanced nodes

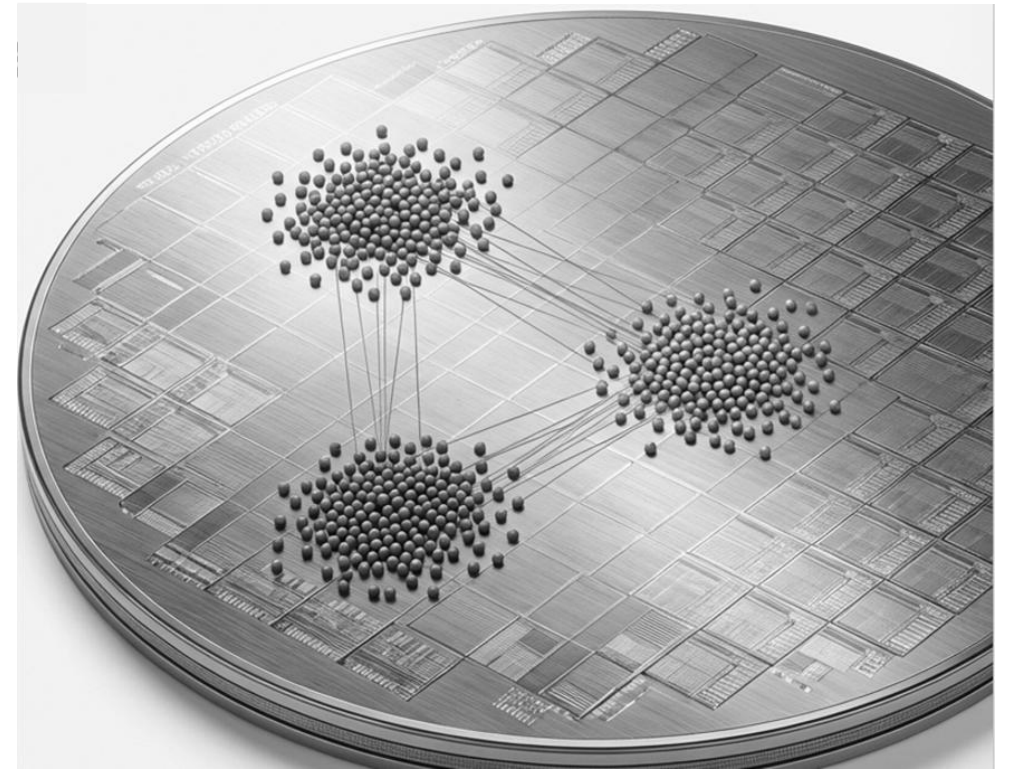


Compute Pipelines

- GPU-accelerated simulation (NVIDIA cuLitho for lithography)
- Scalable ML Ops for model training, inference, and agent orchestration
- Real-time streaming analytics for FDC, APC, and SPC dashboards

Need for an AI-ready data fabric that can reliably ingest, manage, index, version, and retrieve this heterogenous data for AI integrating seamlessly in existing frameworks

Wafer Yield Inspection AI Agent Demo



Wafer Yield Inspection AI Agent for Data Mining & Visualization

• Problem Context

- 1000+ process steps per wafer, TB of data daily
- Complex interactions b/w parameters
- \$5000+ per wafer cost
- Months for root cause analysis

• AI Agent

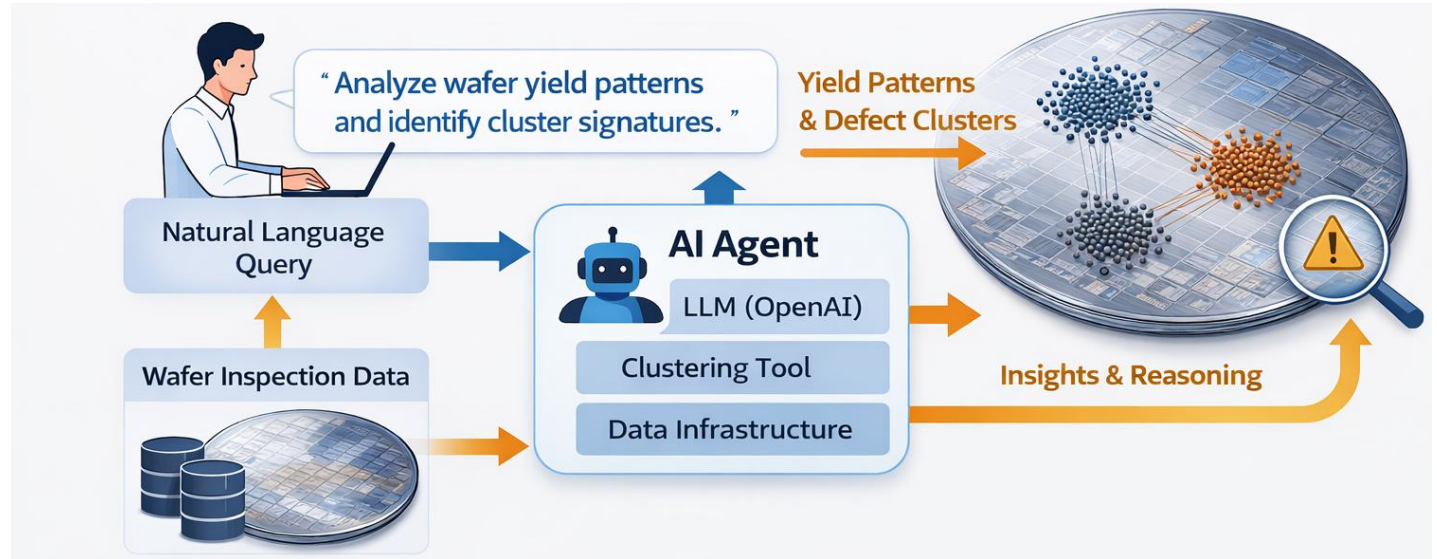
- Plans → Reasons → Uses Tools → Learns & adapts

• AI agent benefits

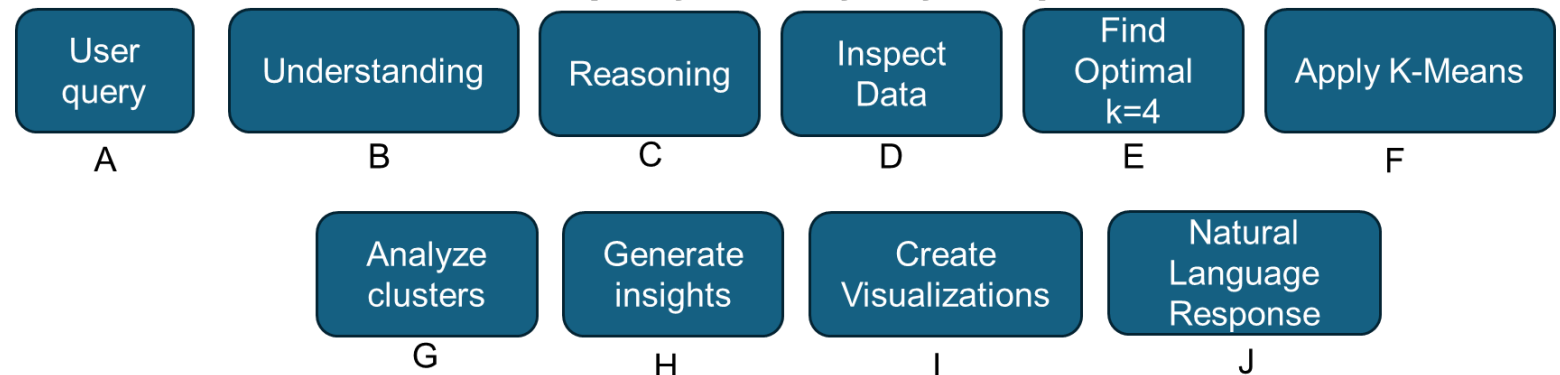
- Reduces analysis time from weeks to minutes
- Handles multi-dimensional data naturally
- Provides explainable insights
- Democratizes advanced analytics
- Scales across fabs

• Demonstration

- How an agent breakdown tasks?
- How tool use (ML + clustering) integrates
- Why agentic loop over rigid pipeline?



Data Flow: Ex query: "Analyze yield patterns"



- **Code Base**
- **Demo video**

Wafer Clustering AI Agent

Videos | Library | Loom

Loom | Free Screen & Video Re...

09be29d3d38e4ef72c.gradio.live

☆

Enter passphrase

Semiconductor Wafer Clustering AI Agent

An intelligent assistant for analyzing semiconductor wafer data using advanced clustering techniques. Powered by LangChain and GPT-4.

Setup

Chat Analysis

Visualizations

Help

Step 1: Initialize the Agent

OpenAI API Key

Initialize Agent

Status

Agent initialized successfully!

Step 2: Load Your Data

Upload CSV

Generate Synthetic

Number of Wafers

1000

5000

Generate Synthetic Data

Load Status

1000 synthetic wafers!

JG

	Defect_Density	Temperature	Pressure	Process_Time	Thickness	Resistivity
Rows	1000					
Columns	13					
Features	Yield_%, Defect_Density, Temperature, Pressure, Process_Time...					

Performance, Scalability & Cost



Performance & Scalability: Real-World Considerations

- **Performance Metrics**

- Query Response: 5-30 seconds
- Clustering 10k wafers: <2 seconds
- Visualization: Real-time
- Token Usage: ~1000-3000 per query

- **Scalability Strategies**

- Caching: Store computed results
- Batching: Process multiple lots
- Async Processing: Non-blocking UI
- Local Models: For sensitive data
- Distributed Computing: For fab-scale

- **Cost Optimization**

- GPT-4: ~\$0.03 per analysis
- Cache frequent queries
- Use GPT-3.5 for simple tasks

Advanced Extensions: Taking It Further

- **Domain-Specific Tools**

- “Spatial defect analysis”, “ML-based yield forecasting”

- **Multi-Modal Analysis**

- Wafer Maps (visual), Time series data, Equipment logs, Metrology Images

- **Integration Capabilities**

- MES systems, Real-time monitoring, Alert generation, Report automation

- **Advanced Reasoning**

- Multi-agent collaboration, Hypothesis generation, Experiment design

Roadmap & Actionable Strategies



Actionable Strategies for the Silicon Lifecycle

01

Start with Assistive AI (L1–L2)

Deploy copilots for documentation search, script generation, and design exploration. Early adopters report 30% faster engineer onboarding and 35% productivity gains in formal verification.

02

Build the Semantic Foundation

Create domain-specific knowledge graphs and ontologies before deploying agents. This prevents LLM hallucinations and grounds reasoning in fab reality (lot → wafer → die → package hierarchy).

03

Instrument for Agent-Ready Data

Ensure storage pipelines support versioning, lineage tracking, and real-time streaming. Agentic systems need reproducible data access to execute autonomous workflows reliably.

04

Adopt Multi-Agent Incrementally

Graduate from single-purpose agents to orchestrated multi-agent systems (L3+). Focus on high-ROI bottlenecks first: yield triage, verification signoff, and DFM checks.

Key Takeaway

Start with assistive AI and semantic data foundations then scale toward multi-agent orchestration across the full design-to-fab pipeline.

The future of silicon is being shaped by autonomous, intelligent agents, the question is no longer whether, but how fast!

Thank You

